

Router Configuration CLI -- Guided Notes ANSWER KEY

Unit 2.4.2 | CompTIA Network+ N10-009 Obj. 2.4 | N10-008 2.4

For instructor use / distribute after students complete the notes.

Question 1

Cisco IOS has three primary privilege levels used in day-to-day work: _____ mode (read-only, prompt ends in >), _____ mode (full access, prompt ends in #), and _____ mode (used to make configuration changes, prompt shows router(config)#).

Answer: User EXEC mode; Privileged EXEC mode; Global Configuration mode.

Question 2

To assign an IP address to a router interface, a technician enters interface configuration mode with _____, then types _____ followed by the IP address and subnet mask, then enables the interface with _____.

Answer: interface [type/number] (e.g., interface GigabitEthernet0/0); ip address [address] [mask]; no shutdown.

Question 3

The command _____ displays the current routing table in Cisco IOS. The letter codes at the start of each route indicate how it was learned: C = _____, S = _____, O = _____.

Answer: show ip route; C = Connected; S = Static; O = OSPF.

Question 4

Running configuration is stored in _____. Startup configuration is stored in _____. If a router reboots without saving, changes made since the last save are _____. The command to save is _____.

Answer: RAM (running-config); NVRAM (startup-config); lost; copy running-config startup-config (or write memory).

Question 5

To configure a static default route on a Cisco router that sends all unmatched traffic to a next-hop address of 10.0.0.1, a technician would type: _____

Answer: ip route 0.0.0.0 0.0.0.0 10.0.0.1

Question 6

The _____ command displays active interfaces, their IP addresses, and their status. An interface showing administratively down means it was disabled with the _____ command. An interface showing down/down indicates a _____ problem.

Answer: show ip interface brief; shutdown; physical (Layer 1) problem.

Question 7

Describe the purpose of the enable secret command versus the enable password command. Which takes precedence when both are configured, and why is one more secure than the other?

Answer: Both set the password for entering privileged EXEC mode. enable secret takes precedence when both are configured. enable secret is more secure because it stores the password as an MD5 hash; enable password stores it in plaintext, visible in show running-config output.

Question 8 -- Real World Application

REAL WORLD: A technician configures a new router interface with an IP address and types no shutdown, but show ip interface brief still shows the interface as down/down. List three physical-layer causes the technician should investigate and explain what command output would help diagnose each one.

Answer: 1) Bad or missing cable -- check show interfaces for input/output errors and line protocol. 2) Speed/duplex mismatch -- check show interfaces for late collisions or excessive errors. 3) Wrong cable type or SFP module mismatch -- physical inspection and show interfaces for transceiver info. Down/down means the physical signal is not present.